

Setup

Architecture:

- Input layer: 3 neurons
- Hidden layer 1: 2 neurons
- Hidden layer 2: 2 neurons
- Output layer: 1 neuron

Input: ["cat", "sat"] **Target:** Y = 1 (positive)

Word embeddings:

"cat" = [1.0, 0.5, 0.2]

"sat" = [0.8, 0.3, 0.6]

Weights (Pre-initialized)

Layer 1:

$W_x^{(1)} = \begin{bmatrix} 0.5 & 0.3 \\ 0.2 & 0.4 \\ 0.1 & 0.6 \end{bmatrix} \quad (3 \times 2)$

$W_h^{(1)} = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix} \quad (2 \times 2)$

$b^{(1)} = [0.1, 0.1] \quad (2 \times 1)$

Layer 2:

$W_{in}^{(2)} = \begin{bmatrix} 0.4 & 0.6 \\ 0.5 & 0.3 \end{bmatrix} \quad (2 \times 2)$

$W_h^{(2)} = \begin{bmatrix} 0.6 & 0.4 \\ 0.2 & 0.7 \end{bmatrix} \quad (2 \times 2)$

$b^{(2)} = [0.1, 0.1] \quad (2 \times 1)$

Output Layer:

$$W_y = \begin{bmatrix} 0.9 \\ 0.4 \end{bmatrix} \quad (2 \times 1)$$

$$b_y = [0.2] \quad (1 \times 1)$$

FORWARD PROPAGATION

Initialization (t=0)

$$h_0^{(1)} = [0, 0]$$

$$h_0^{(2)} = [0, 0]$$

Token 1: "cat" (t=1)

Step 1: Process Layer 1

Input:

$$x_1 = [1.0, 0.5, 0.2]$$

Compute weighted sum:

$$z_1^{(1)} = W_x^{(1)} \cdot x_1 + W_h^{(1)} \cdot h_0^{(1)} + b^{(1)}$$

$$W_x^{(1)} \cdot x_1 = \begin{bmatrix} 0.5 & 0.3 \\ 0.2 & 0.4 \end{bmatrix} \cdot \begin{bmatrix} 1.0 \\ 0.5 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.5 \times 1.0 + 0.3 \times 0.5 + 0.1 \times 0.2 \\ 0.2 \times 1.0 + 0.4 \times 0.5 + 0.6 \times 0.2 \end{bmatrix} = \begin{bmatrix} 0.67 \\ 0.52 \end{bmatrix}$$

$$\begin{bmatrix} 0.2 & 0.4 \end{bmatrix} \times \begin{bmatrix} 0.5 \end{bmatrix} = \begin{bmatrix} 0.2 \times 1.0 + 0.4 \times 0.5 + 0.6 \times 0.2 \end{bmatrix} = \begin{bmatrix} 0.52 \end{bmatrix}$$

$$\begin{bmatrix} 0.1 & 0.6 \end{bmatrix} \cdot \begin{bmatrix} 0.2 \end{bmatrix}$$

$$W_h^{(1)} \cdot h_0^{(1)} = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0.3 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0 \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

$$z_1^{(1)} = \begin{bmatrix} 0.67 \\ 0.52 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.62 \end{bmatrix} = \begin{bmatrix} 0.77 \\ 1.14 \end{bmatrix}$$

$$\begin{bmatrix} 0.52 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0.1 \\ 0.62 \end{bmatrix}$$

Apply activation (tanh):

$$h_1^{(1)} = \tanh(z_1^{(1)}) = \tanh([0.77, 0.62])$$

$$= [0.646, 0.551]$$

Step 2: Process Layer 2

Input from Layer 1:

$$h_1^{(1)} = [0.646, 0.551]$$

Compute weighted sum:

$$z_1^{(2)} = W_{in}^{(2)} \cdot h_1^{(1)} + W_h^{(2)} \cdot h_0^{(2)} + b^{(2)}$$

$$W_{in}^{(2)} \cdot h_1^{(1)} = [[0.4, 0.6], [0.646] \quad [0.4 \times 0.646 + 0.6 \times 0.551] \quad [0.589]$$

$$[0.5, 0.3]] \times [0.551] = [0.5 \times 0.646 + 0.3 \times 0.551] = [0.488]$$

$$W_h^{(2)} \cdot h_0^{(2)} = [[0.6, 0.4], [0] \quad [0]$$

$$[0.2, 0.7]] \times [0] = [0]$$

$$z_1^{(2)} = [0.589] + [0] + [0.1] = [0.689]$$

$$[0.488] \quad [0] \quad [0.1] \quad [0.588]$$

Apply activation (tanh):

$$h_1^{(2)} = \tanh(z_1^{(2)}) = \tanh([0.689, 0.588])$$

$$= [0.597, 0.528]$$

Step 3: Compute Output y_1

Input from Layer 2:

$$h_1^{(2)} = [0.597, 0.528]$$

Compute output:

$$z_1 = W_y \cdot h_1^{(2)} + b_y$$

$$W_y \cdot h_1^{(2)} = [[0.9], [0.597] \quad [0.9 \times 0.597 + 0.4 \times 0.528] \quad [0.748]$$

$$[0.4]] \times [0.528] =$$

$$z_1 = [0.748] + [0.2] = [0.948]$$

Apply sigmoid (for binary classification):

$$y_1 = \text{sigmoid}(0.948) = 1/(1 + e^{(-0.948)}) = 0.721$$

Summary at t=1:

$$x_1 = [1.0, 0.5, 0.2]$$

$$h_1^{(1)} = [0.646, 0.551]$$

$$h_1^{(2)} = [0.597, 0.528]$$

$$y_1 = 0.721$$

Note: We don't calculate loss yet for many-to-one!

Token 2: "sat" (t=2)

Step 1: Process Layer 1

Input:

$$x_2 = [0.8, 0.3, 0.6]$$

Compute weighted sum (NOW using $h_1^{(1)}$):

$$z_2^{(1)} = W_x^{(1)} \cdot x_2 + W_h^{(1)} \cdot h_1^{(1)} + b^{(1)}$$

$$W_x^{(1)} \cdot x_2 = \begin{bmatrix} 0.5 & 0.3 \\ 0.8 \end{bmatrix} \begin{bmatrix} 0.5 \times 0.8 + 0.3 \times 0.3 + 0.1 \times 0.6 \end{bmatrix} \begin{bmatrix} 0.55 \end{bmatrix}$$

$$\begin{bmatrix} 0.2 & 0.4 \end{bmatrix} \times \begin{bmatrix} 0.3 \end{bmatrix} = \begin{bmatrix} 0.2 \times 0.8 + 0.4 \times 0.3 + 0.6 \times 0.6 \end{bmatrix} = \begin{bmatrix} 0.64 \end{bmatrix}$$

$$\begin{bmatrix} 0.1 & 0.6 \end{bmatrix} \begin{bmatrix} 0.6 \end{bmatrix}$$

$$W_h^{(1)} \cdot h_1^{(1)} = \begin{bmatrix} 0.7 & 0.2 \end{bmatrix} \begin{bmatrix} 0.646 \end{bmatrix} \begin{bmatrix} 0.7 \times 0.646 + 0.2 \times 0.551 \end{bmatrix} \begin{bmatrix} 0.562 \end{bmatrix}$$

$$\begin{bmatrix} 0.3 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.551 \end{bmatrix} = \begin{bmatrix} 0.3 \times 0.646 + 0.8 \times 0.551 \end{bmatrix} = \begin{bmatrix} 0.635 \end{bmatrix}$$

↑

This is the recurrent connection! Memory from token 1!

$$z_2^{(1)} = [0.55] + [0.562] + [0.1] = [1.212]$$

[0.64] [0.635] [0.1] [1.375]

Apply activation:

$$h_2^{(1)} = \tanh([1.212, 1.375]) = [0.837, 0.878]$$

Step 2: Process Layer 2

Input from Layer 1:

$$h_2^{(1)} = [0.837, 0.878]$$

Compute weighted sum (NOW using $h_1^{(2)}$):

$$z_2^{(2)} = W_{in}^{(2)} \cdot h_2^{(1)} + W_h^{(2)} \cdot h_1^{(2)} + b^{(2)}$$

$$W_{in}^{(2)} \cdot h_2^{(1)} = [[0.4, 0.6], [0.837] \quad [0.4 \times 0.837 + 0.6 \times 0.878] \quad [0.862]$$

$$[0.5, 0.3]] \times [0.878] = [0.5 \times 0.837 + 0.3 \times 0.878] = [0.682]$$

$$W_h^{(2)} \cdot h_1^{(2)} = [[0.6, 0.4], [0.597] \quad [0.6 \times 0.597 + 0.4 \times 0.528] \quad [0.569]$$

$$[0.2, 0.7]] \times [0.528] = [0.2 \times 0.597 + 0.7 \times 0.528] = [0.489]$$

↑

Recurrent connection from previous timestamp!

$$z_2^{(2)} = [0.862] + [0.569] + [0.1] = [1.531]$$

$$[0.682] \quad [0.489] \quad [0.1] \quad [1.271]$$

Apply activation:

$$h_2^{(2)} = \tanh([1.531, 1.271]) = [0.912, 0.852]$$

Step 3: Compute Output y_2 (FINAL OUTPUT)

Input from Layer 2:

$$h_2^{(2)} = [0.912, 0.852]$$

Compute output:

$$z_2 = W_y \cdot h_2^{(2)} + b_y$$

$$W_y \cdot h_2^{(2)} = [[0.9], \quad [0.912] \quad [0.9 \times 0.912 + 0.4 \times 0.852] \quad [1.162] \\ [0.4]] \times [0.852] =$$

$$z_2 = [1.162] + [0.2] = [1.362]$$

Apply sigmoid:

$$y_2 = \text{sigmoid}(1.362) = 1/(1 + e^{(-1.362)}) = 0.796$$

Summary at t=2:

$$x_2 = [0.8, 0.3, 0.6]$$

$$h_2^{(1)} = [0.837, 0.878]$$

$$h_2^{(2)} = [0.912, 0.852]$$

$$y_2 = 0.796 \leftarrow \text{This is our FINAL prediction!}$$

LOSS CALCULATION

Calculate Loss (Only at Final Timestamp)

For many-to-one, we only use the final output:

$$\text{Predicted: } y_2 = 0.796$$

$$\text{True label: } Y = 1 \text{ (positive)}$$

Binary Cross-Entropy Loss:

$$\text{Loss} = -[Y \times \log(y_2) + (1-Y) \times \log(1-y_2)]$$

$$\text{Loss} = -[1 \times \log(0.796) + 0 \times \log(0.204)]$$

$$= -[1 \times (-0.228) + 0]$$

$$= -[-0.228]$$

$$= 0.228$$

Complete Forward Pass Summary

Token 1: "cat"
$x_1 = [1.0, 0.5, 0.2]$
$h_1^{(1)} = [0.646, 0.551]$
$h_1^{(2)} = [0.597, 0.528]$
$y_1 = 0.721$ (ignored in loss)

Token 2: "sat"
$x_2 = [0.8, 0.3, 0.6]$
$h_2^{(1)} = [0.837, 0.878] \leftarrow \text{uses } h_1^{(1)}$
$h_2^{(2)} = [0.912, 0.852] \leftarrow \text{uses } h_1^{(2)}$
$y_2 = 0.796 \leftarrow \text{FINAL PREDICTION}$

Loss Calculation
$\text{Loss} = \text{CrossEntropy}(y_2, Y)$
$= 0.228$

Key Points

- ✓ **Forward prop happens sequentially** through all tokens
- ✓ **Each layer processes in order:** Input $\rightarrow$ Layer 1 $\rightarrow$ Layer 2 $\rightarrow$ Output
- ✓ **Recurrent connections carry information:**
 - $h_1^{(1)} \rightarrow h_2^{(1)}$

- $h_1^{(2)} \rightarrow h_2^{(2)}$

✓ **Loss calculated only once** at the final token (for many-to-one)

✓ **Intermediate outputs (y_1) are ignored** in many-to-one tasks